| No. | Status | Date |
|---|---|---|
| LNDE-RPT-96-014-DS-6-SG-101 | Issued | 1996-03-08 |

# UT Thickness Data Sheet - 6-SG-101

## Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for the sour gas inlet line 6-SG-101 located within the ATU-100. The objective is to establish baseline wall thickness data for the asset prior to the unit entering service.

Grace Okafor, acting as the UT technician, performed the measurements and data recording. All activities were conducted in accordance with the requirements of the SNT-TC-1A UT Level II certification.

## Technical Basis

The thickness measurements were performed under QSCP-NDT-001 Rev 0. The inspection methodology and instrument calibration were executed in accordance with ASTM E797.

The results recorded herein are integrated into the baseline survey documented in LNDE-RPT-96-014, which covers 118 CMLs for the ATU-100.

## Line Specifications

The subject line 6-SG-101 is constructed from A106 Gr.B with an outer diameter of 168.3 mm and a nominal wall thickness of 7.11 mm. The design pressure is 1965 kPa(g) and the design temperature is 66 °C.

The minimum allowable thickness (t-min) for this line is 3.56 mm. Any reading below this threshold is subject to an NCR.

## Inspection Results

Thickness readings were taken at the designated locations for 6-SG-101. As the unit is not yet in service, these values serve as the initial baseline for future corrosion rate and remaining life calculations.

The measured values are compared against the t-min of 3.56 mm to ensure the line meets the required integrity standards for the ATU-100 pre-startup phase.

## Thickness readings, measured 1996-03-08

**Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-SG-101-01 | 7.11 | 3.56 | — | — | 7.12 | — | — | — | 2001-03-08 | |
| 6-SG-101-02 | 7.11 | 3.56 | — | — | 7.05 | — | — | — | 2001-03-08 | |
| 6-SG-101-03 | 7.11 | 3.56 | — | — | 7.08 | — | — | — | 2001-03-08 | |
| 6-SG-101-04 | 7.11 | 3.56 | — | — | 7.09 | — | — | — | 2001-03-08 | |
| 6-SG-101-05 | 7.11 | 3.56 | — | — | 7.14 | — | — | — | 2001-03-08 | |
| 6-SG-101-06 | 7.11 | 3.56 | — | — | 7.07 | — | — | — | 2001-03-08 | |

**Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-SG-102-01 | 7.11 | 3.56 | — | — | 7.14 | — | — | — | 2001-03-08 | |
| 6-SG-102-02 | 7.11 | 3.56 | — | — | 7.11 | — | — | — | 2001-03-08 | |
| 6-SG-102-03 | 7.11 | 3.56 | — | — | 7.04 | — | — | — | 2001-03-08 | |
| 6-SG-102-04 | 7.11 | 3.56 | — | — | 6.99 | — | — | — | 2001-03-08 | |
| 6-SG-102-05 | 7.11 | 3.56 | — | — | 7.13 | — | — | — | 2001-03-08 | |

**Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 8-TG-101-01 | 8.18 | 4.06 | — | — | 8.20 | — | — | — | 2006-03-08 | |
| 8-TG-101-02 | 8.18 | 4.06 | — | — | 8.20 | — | — | — | 2006-03-08 | |
| 8-TG-101-03 | 8.18 | 4.06 | — | — | 8.23 | — | — | — | 2006-03-08 | |

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 8-TG-101-04 | 8.18 | 4.06 | — | — | 8.10 | — | — | — | 2006-03-08 | |
| 8-TG-101-05 | 8.18 | 4.06 | — | — | 8.22 | — | — | — | 2006-03-08 | |

**Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-101-01 | 6.02 | 3.18 | — | — | 5.99 | — | — | — | 2006-03-08 | |
| 4-LA-101-02 | 6.02 | 3.18 | — | — | 6.01 | — | — | — | 2006-03-08 | |
| 4-LA-101-03 | 6.02 | 3.18 | — | — | 5.96 | — | — | — | 2006-03-08 | |
| 4-LA-101-04 | 6.02 | 3.18 | — | — | 6.07 | — | — | — | 2006-03-08 | |
| 4-LA-101-05 | 6.02 | 3.18 | — | — | 6.01 | — | — | — | 2006-03-08 | |
| 4-LA-101-06 | 6.02 | 3.18 | — | — | 6.06 | — | — | — | 2006-03-08 | |
| 4-LA-101-07 | 6.02 | 3.18 | — | — | 5.99 | — | — | — | 2006-03-08 | |

**Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-102-01 | 6.02 | 3.18 | — | — | 6.02 | — | — | — | 2006-03-08 | |
| 4-LA-102-02 | 6.02 | 3.18 | — | — | 6.07 | — | — | — | 2006-03-08 | |
| 4-LA-102-03 | 6.02 | 3.18 | — | — | 6.07 | — | — | — | 2006-03-08 | |
| 4-LA-102-04 | 6.02 | 3.18 | — | — | 5.99 | — | — | — | 2006-03-08 | |
| 4-LA-102-05 | 6.02 | 3.18 | — | — | 6.00 | — | — | — | 2006-03-08 | |
| 4-LA-102-06 | 6.02 | 3.18 | — | — | 6.00 | — | — | — | 2006-03-08 | |
| 4-LA-102-07 | 6.02 | 3.18 | — | — | 6.00 | — | — | — | 2006-03-08 | |

**Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-LA-103-01 | 7.11 | 3.56 | — | — | 7.02 | — | — | — | 2006-03-08 | |
| 6-LA-103-02 | 7.11 | 3.56 | — | — | 7.19 | — | — | — | 2006-03-08 | |
| 6-LA-103-03 | 7.11 | 3.56 | — | — | 7.08 | — | — | — | 2006-03-08 | |
| 6-LA-103-04 | 7.11 | 3.56 | — | — | 7.17 | — | — | — | 2006-03-08 | |
| 6-LA-103-05 | 7.11 | 3.56 | — | — | 7.19 | — | — | — | 2006-03-08 | |
| 6-LA-103-06 | 7.11 | 3.56 | — | — | 7.16 | — | — | — | 2006-03-08 | |
| 6-LA-103-07 | 7.11 | 3.56 | — | — | 7.18 | — | — | — | 2006-03-08 | |
| 6-LA-103-08 | 7.11 | 3.56 | — | — | 7.04 | — | — | — | 2006-03-08 | |

**Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-101-01 | 8.56 | 3.81 | — | — | 8.52 | — | — | — | 2006-03-08 | |
| 4-RA-101-02 | 8.56 | 3.81 | — | — | 8.54 | — | — | — | 2006-03-08 | |
| 4-RA-101-03 | 8.56 | 3.81 | — | — | 8.61 | — | — | — | 2006-03-08 | |
| 4-RA-101-04 | 8.56 | 3.81 | — | — | 8.63 | — | — | — | 2006-03-08 | |
| 4-RA-101-05 | 8.56 | 3.81 | — | — | 8.60 | — | — | — | 2006-03-08 | |
| 4-RA-101-07 | 8.56 | 3.81 | — | — | 8.60 | — | — | — | 2006-03-08 | |
| 4-RA-101-08 | 8.56 | 3.81 | — | — | 8.68 | — | — | — | 2006-03-08 | |
| 4-RA-101-09 | 8.56 | 3.81 | — | — | 8.57 | — | — | — | 2006-03-08 | |
| 4-RA-101-10 | 8.56 | 3.81 | — | — | 8.67 | — | — | — | 2006-03-08 | |
| 4-RA-101-11 | 8.56 | 3.81 | — | — | 8.68 | — | — | — | 2006-03-08 | |

**Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-102-01 | 8.56 | 3.81 | — | — | 8.54 | — | — | — | 2006-03-08 | |
| 4-RA-102-02 | 8.56 | 3.81 | — | — | 8.57 | — | — | — | 2006-03-08 | |
| 4-RA-102-03 | 8.56 | 3.81 | — | — | 8.56 | — | — | — | 2006-03-08 | |
| 4-RA-102-04 | 8.56 | 3.81 | — | — | 8.54 | — | — | — | 2006-03-08 | |
| 4-RA-102-05 | 8.56 | 3.81 | — | — | 8.55 | — | — | — | 2006-03-08 | |
| 4-RA-102-06 | 8.56 | 3.81 | — | — | 8.51 | — | — | — | 2006-03-08 | |
| 4-RA-102-07 | 8.56 | 3.81 | — | — | 8.58 | — | — | — | 2006-03-08 | |
| 4-RA-102-08 | 8.56 | 3.81 | — | — | 8.49 | — | — | — | 2006-03-08 | |
| 4-RA-102-09 | 8.56 | 3.81 | — | — | 8.55 | — | — | — | 2006-03-08 | |

**Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-RA-103-01 | 10.97 | 4.57 | — | — | 11.00 | — | — | — | 2006-03-08 | |
| 6-RA-103-02 | 10.97 | 4.57 | — | — | 11.09 | — | — | — | 2006-03-08 | |
| 6-RA-103-03 | 10.97 | 4.57 | — | — | 10.99 | — | — | — | 2006-03-08 | |
| 6-RA-103-04 | 10.97 | 4.57 | — | — | 10.98 | — | — | — | 2006-03-08 | |
| 6-RA-103-05 | 10.97 | 4.57 | — | — | 10.95 | — | — | — | 2006-03-08 | |
| 6-RA-103-06 | 10.97 | 4.57 | — | — | 11.01 | — | — | — | 2006-03-08 | |
| 6-RA-103-07 | 10.97 | 4.57 | — | — | 10.99 | — | — | — | 2006-03-08 | |
| 6-RA-103-08 | 10.97 | 4.57 | — | — | 10.97 | — | — | — | 2006-03-08 | |
| 6-RA-103-09 | 10.97 | 4.57 | — | — | 11.01 | — | — | — | 2006-03-08 | |
| 6-RA-103-10 | 10.97 | 4.57 | — | — | 11.02 | — | — | — | 2006-03-08 | |

**Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-RV-101-01 | 7.62 | 3.56 | — | — | 7.59 | — | — | — | 2001-03-08 | |
| 3-RV-101-02 | 7.62 | 3.56 | — | — | 7.57 | — | — | — | 2001-03-08 | |
| 3-RV-101-03 | 7.62 | 3.56 | — | — | 7.52 | — | — | — | 2001-03-08 | |
| 3-RV-101-04 | 7.62 | 3.56 | — | — | 7.59 | — | — | — | 2001-03-08 | |
| 3-RV-101-05 | 7.62 | 3.56 | — | — | 7.67 | — | — | — | 2001-03-08 | |
| 3-RV-101-06 | 7.62 | 3.56 | — | — | 7.59 | — | — | — | 2001-03-08 | |

**Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RV-102-01 | 5.54 | 2.79 | — | — | 5.53 | — | — | — | 2001-03-08 | |
| 2-RV-102-02 | 5.54 | 2.79 | — | — | 5.59 | — | — | — | 2001-03-08 | |
| 2-RV-102-03 | 5.54 | 2.79 | — | — | 5.48 | — | — | — | 2001-03-08 | |
| 2-RV-102-04 | 5.54 | 2.79 | — | — | 5.58 | — | — | — | 2001-03-08 | |

**Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-RF-101-01 | 5.49 | 2.41 | — | — | 5.48 | — | — | — | 2006-03-08 | |
| 3-RF-101-02 | 5.49 | 2.41 | — | — | 5.45 | — | — | — | 2006-03-08 | |
| 3-RF-101-03 | 5.49 | 2.41 | — | — | 5.45 | — | — | — | 2006-03-08 | |
| 3-RF-101-04 | 5.49 | 2.41 | — | — | 5.46 | — | — | — | 2006-03-08 | |
| 3-RF-101-05 | 5.49 | 2.41 | — | — | 5.45 | — | — | — | 2006-03-08 | |
| 3-RF-101-06 | 5.49 | 2.41 | — | — | 5.52 | — | — | — | 2006-03-08 | |
| 3-RF-101-07 | 5.49 | 2.41 | — | — | 5.42 | — | — | — | 2006-03-08 | |

**Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RF-102-01 | 3.91 | 2.16 | — | — | 3.92 | — | — | — | 2006-03-08 | |
| 2-RF-102-02 | 3.91 | 2.16 | — | — | 3.88 | — | — | — | 2006-03-08 | |
| 2-RF-102-03 | 3.91 | 2.16 | — | — | 3.93 | — | — | — | 2006-03-08 | |
| 2-RF-102-04 | 3.91 | 2.16 | — | — | 3.87 | — | — | — | 2006-03-08 | |
| 2-RF-102-07 | 3.91 | 2.16 | — | — | 3.96 | — | — | — | 2006-03-08 | |

**Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-104-01 | 6.02 | 3.18 | — | — | 6.07 | — | — | — | 2006-03-08 | |
| 4-LA-104-02 | 6.02 | 3.18 | — | — | 6.03 | — | — | — | 2006-03-08 | |
| 4-LA-104-03 | 6.02 | 3.18 | — | — | 5.96 | — | — | — | 2006-03-08 | |
| 4-LA-104-04 | 6.02 | 3.18 | — | — | 6.06 | — | — | — | 2006-03-08 | |
| 4-LA-104-05 | 6.02 | 3.18 | — | — | 6.04 | — | — | — | 2006-03-08 | |
| 4-LA-104-06 | 6.02 | 3.18 | — | — | 6.10 | — | — | — | 2006-03-08 | |
| 4-LA-104-07 | 6.02 | 3.18 | — | — | 6.01 | — | — | — | 2006-03-08 | |
| 4-LA-104-08 | 6.02 | 3.18 | — | — | 5.94 | — | — | — | 2006-03-08 | |

**Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-104-01 | 8.56 | 3.81 | — | — | 8.51 | — | — | — | 2006-03-08 | |
| 4-RA-104-02 | 8.56 | 3.81 | — | — | 8.55 | — | — | — | 2006-03-08 | |
| 4-RA-104-03 | 8.56 | 3.81 | — | — | 8.54 | — | — | — | 2006-03-08 | |
| 4-RA-104-04 | 8.56 | 3.81 | — | — | 8.59 | — | — | — | 2006-03-08 | |
| 4-RA-104-05 | 8.56 | 3.81 | — | — | 8.60 | — | — | — | 2006-03-08 | |
| 4-RA-104-06 | 8.56 | 3.81 | — | — | 8.59 | — | — | — | 2006-03-08 | |
| 4-RA-104-07 | 8.56 | 3.81 | — | — | 8.59 | — | — | — | 2006-03-08 | |

**Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-ST-101-01 | 3.91 | 2.29 | — | — | 3.93 | — | — | — | 2006-03-08 | |
| 2-ST-101-02 | 3.91 | 2.29 | — | — | 3.90 | — | — | — | 2006-03-08 | |
| 2-ST-101-03 | 3.91 | 2.29 | — | — | 3.94 | — | — | — | 2006-03-08 | |
| 2-ST-101-04 | 3.91 | 2.29 | — | — | 3.82 | — | — | — | 2006-03-08 | |
| 2-ST-101-05 | 3.91 | 2.29 | — | — | 3.91 | — | — | — | 2006-03-08 | |

**Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-CW-101-01 | 5.49 | 2.79 | — | — | 5.42 | — | — | — | 2006-03-08 | |
| 3-CW-101-02 | 5.49 | 2.79 | — | — | 5.41 | — | — | — | 2006-03-08 | |
| 3-CW-101-03 | 5.49 | 2.79 | — | — | 5.53 | — | — | — | 2006-03-08 | |
| 3-CW-101-04 | 5.49 | 2.79 | — | — | 5.46 | — | — | — | 2006-03-08 | |
| 3-CW-101-05 | 5.49 | 2.79 | — | — | 5.48 | — | — | — | 2006-03-08 | |

**Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-CW-102-01 | 5.49 | 2.79 | — | — | 5.50 | — | — | — | 2006-03-08 | |
| 3-CW-102-02 | 5.49 | 2.79 | — | — | 5.51 | — | — | — | 2006-03-08 | |
| 3-CW-102-03 | 5.49 | 2.79 | — | — | 5.53 | — | — | — | 2006-03-08 | |
| 3-CW-102-04 | 5.49 | 2.79 | — | — | 5.51 | — | — | — | 2006-03-08 | |

| Sign-off | Name | Function | Date | Signature |
|---|---|---|---|---|
| Prepared | Grace Okafor | UT technician, SNT-TC-1A UT Level II | 1996-03-08 | _____ |